

OPERATING MANUAL

MAZATROL SmoothAi
MAZATROL SmoothG
MAZATROL SmoothG³

Ai THERMAL SHIELD

Manual No.: H747SA0025E

Serial No.:

Before using this machine and equipment, fully understand the contents of this manual to ensure proper operation. Should any questions arise, please ask the nearest Technical Center or Technology Center.

IMPORTANT NOTICE

1. Be sure to observe the safety precautions described in this manual and the contents of the safety plates on the machine and equipment. Failure may cause serious personal injury or material damage. Please replace any missing safety plates as soon as possible.
2. No modifications are to be performed that will affect operation safety.
3. For the purpose of explaining the operation of the machine and equipment, some illustrations may not include safety features such as covers, doors, etc. Before operation, make sure all such items are in place.
4. If there is a difference between the contents of the manual and the actual state of the machine, please contact the nearest Technical Center or Technology Center.
5. Always keep this manual near the machinery for immediate use.
6. If a new manual is required, please order from the nearest Technical Center or Technology Center with the manual No. or the machine name, serial No. and manual name.
7. The export of machining programs may be restricted to prevent them from being used for military purposes. The machining programs you prepared are your property, so you should manage them yourself in accordance with the law.
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SAFETY PRECAUTIONS

This manual is prepared only for the function or the device which is specified in the title of this manual. Therefore, you are required to carefully read and understand the operating manual of the machine, especially the section describing safety precautions, before using or operating the function or the device.

1 INTRODUCTION

An object generally expands and contracts as the temperature rises and drops, respectively. For example, a 1 m-long iron product expands about 10 μm if its temperature rises by 1°C.

Since the temperatures of the structural members of the machine change in response to the ambient temperature and to the heat generated by machine operation, an unexpected positional error is produced between the workpiece and the tool tip.

The function Ai Thermal Shield is designed to compensate for such an error to enhance machining accuracy using the following two methods:

<Compensation factor adjustment>

- The change in machine temperature and the amounts of compensation for thermal displacement are shown graphically on a dedicated display.
- The optimum amount of compensation can be obtained by modifying the compensation factor according to the change in ambient temperature.

<Learning function>

- The temperature change and the error in the entered workpiece dimension are shown on a dedicated display.
- The display shows the dimension error that may be produced when the learning results are applied.
- Through learning, the compensation formulas are optimized according to the environmental temperature change in the customer's factory.
- With this function, an adjustment can be made more effectively compared to the compensation factor adjustment function.

1

INTRODUCTION



Handwriting practice area with horizontal dashed lines.

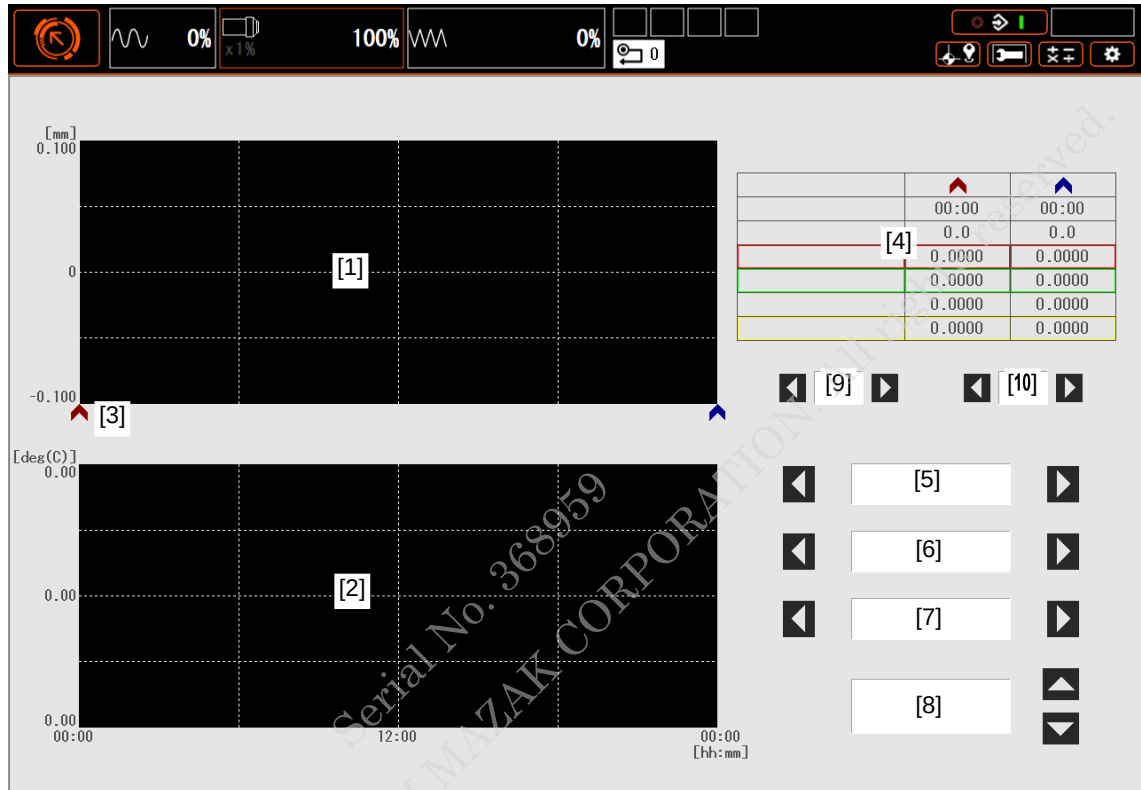
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2 Ai THERMAL SHIELD (COMPENSATION FACTOR ADJUSTMENT)

This function makes it possible to adjust the compensation factor while checking the temperature change and the amount of compensation.

2-1 Data Display

1. Displayed data



D747SA0001

2. Display data description

No.	Item	Unit	Data description
[1]	Compensation amount graph	—	A graph showing the previous compensation value and the value after an adjustment of the compensation factor. Green: Reference compensation amount after adjustment Red: Compensation amount for thermal displacement Yellow: Composite compensation amount Remark : Select the [Show Composite Comp.] menu item to graphically display the composite compensation amount.
[2]	Temperature graph	—	A graph of temperatures. Red: Room temperature (or room temperature 1) Green: Room temperature 2 Orange: Coolant temperature Remark : Use the [Show temp. in Fahrenheit] menu item to display the temperature in Celsius or Fahrenheit. Note : It depends upon the specifications of the machine what kinds of temperature are displayed.
[3]	Slider	—	Used to set the time of the numerical data to be displayed in [4].

No.	Item	Unit	Data description
[4]	Compensation data	—	Compensation data at the time specified by the slider [3]. The difference in compensation amount is calculated from “Reference compensation amount after adjustment” and “Compensation amount for thermal displacement.” Remark : Select the [DIFF MODE] menu item to display the difference between two values specified by the respective sliders. Note : The displayed difference in compensation amount may have an error in the fourth (ten thousandths) decimal place as compared with the internal calculation.
[5]	Date	Y/M/D	Date of the graph to be displayed. The date can be changed with the buttons  and .
[6]	Time	H : M	Starting time of the graph to be displayed. The starting time can be changed with the buttons  and .
[7]	Compensation axis	—	Compensation axis of the graph to be displayed. The axis can be changed with the buttons  and .
[8]	Factor for adjusting the compensation amount	—	Compensation factor after adjustment of the compensation amount for thermal displacement. The factor can be changed with the buttons  and . Standard setting: 1 Setting range: -3.0 to 3.0 (Setting unit = 0.1)
[9]	TNo.	—	Number (TNo.) of the tool for which the composite compensation amount is to be displayed graphically. The tool number can be changed with the buttons  and .
[10]	Axis	—	Axis of the composite compensation amount to be displayed graphically. The axis can be changed with the buttons  and .

2-2 Change of Scales

1. Automatic vertical-axis scaling for the compensation amount graph

Tap the **[COMP. AUTO SCALE]** menu item to highlight the menu item, and the scale of the vertical axis will automatically be changed so as to show the graph of compensation amount in its entirety in an optimum way for the specified date and the next day each time the “Date” is changed.

2. Manual vertical-axis scaling for the compensation amount graph

Follow the procedure below to manually change the scale of the vertical axis for compensation amount as desired.

(1) Tap the **[COMP. SCALE SELECT]** menu item.

(2) Enter the highest value of the scale.

Setting range: 0.001 to 0.999 mm

→ The graph of compensation amount is redrawn according to the new scale.

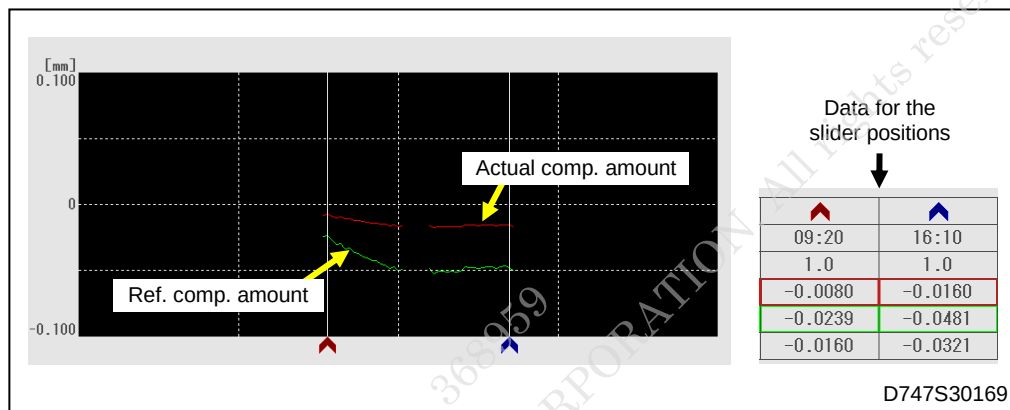
3. Manual horizontal-axis scaling for the compensation amount and the temperature graph

Each tap on the **[TIME SCALE CHANGE]** menu item changes the scale of the horizontal axis for the time of day in rotation (among three sizes of 6, 12 and 24 hours).

2-3 Adjustment of Compensation Factor

Check the information in the graph and follow the procedure below to adjust the compensation factor.

- (1) Change the date of the graph to be displayed.
- (2) Change the starting time of the graph to be displayed.
- (3) Change the compensation axis of the graph to be displayed.
- (4) Change the compensation factor.
 - ➔ The graph of the adjusted reference compensation amount changes according to the change in compensation factor.
 - The factor can be changed by moving the slider while checking the actual compensation data.



- (5) Tap the **[APPLY]** menu item.
 - ➔ The adjusted compensation factor becomes effective.

Note 1: Graphs for the last 3 months (maximum) can be displayed.

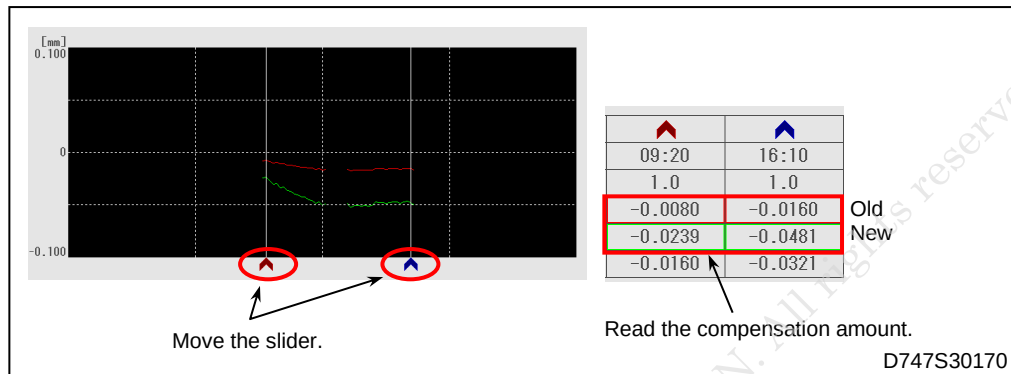
Note 2: The selectable compensation axes depend upon the machine specifications.

2-4 Examples of Use

1. To reduce (to 0) the difference about +0.02 mm in accuracy error (in Z-axis direction) from morning to evening

- (1) Select the date of machining and set the Z-axis as compensation axis.
- (2) Experiment with a new compensation factor.
- (3) Move the slider to the morning and evening positions to read the compensation amounts.

<If '3' is set as the compensation factor>



	Morning (09 : 20)	Evening (16 : 10)	Change in comp. amount
Compensation amount for thermal displacement	-0.0080 mm	-0.0160 mm	-0.0080 mm
Reference compensation amount after adjustment	-0.0239 mm	-0.0481 mm	-0.0242 mm

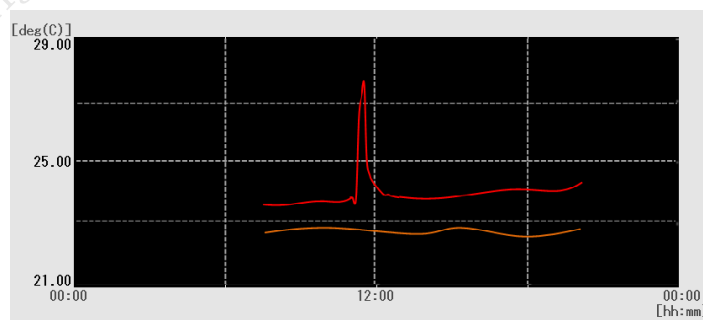
- As shown in the above table, if '3' is set as the compensation factor, the change in the amount of thermal displacement compensation from morning to evening is -0.0242 mm.
- Machining accuracy can be improved for an error of about 0.02 mm since the amount of compensation is changed from -0.0080 mm to -0.0242 mm.

- (4) Tap the **[APPLY]** menu item to make the adjusted compensation factor effective.
- (5) From the next day onward, check to see if machining accuracy is improved.

2. To improve the accuracy of some workpieces

- (1) Select the day on which the workpieces were machined.
- (2) Check the room temperature and coolant temperature graphs.
 - Check if these temperatures changed significantly in the period of time when the workpieces of low accuracy were machined.

<Example of abrupt change in room temperature>



- (3) Find and remove the cause of the temperature change, and check if the accuracy is improved.

2-5 Precautions

1. This function is not designed to guarantee accuracy.
2. An adjustment of the compensation factor involves a change in the amount of thermal displacement compensation and, therefore, has a direct effect on misalignment and machining dimensions. When a workpiece is machined for the first time after a change of compensation factor, due allowance should be made for dimensional errors.
3. The change in compensation factor is used for all tools used for the machining concerned.
4. The amount of compensation set with the thermal displacement compensation function is not incorporated in the POSITION and MACHINE values on the **POSITION** display, which means that a change of compensation factor does not cause any change on the **POSITION** display.
5. When using this function, determine the change in the amount of thermal displacement compensation on a daily basis and adjust the compensation factor for the next day's operation. Adjustments and learning should not be performed many times a day.
6. If Ai Thermal Shield is down due to a wire break in the thermal sensor or for any other reason, do not try to collect data or execute learning, using the learning conditions. If this is the case, contact Mazak Technical Center or Technology Center.



Blank area for notes or adjustments, featuring horizontal dashed lines.

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3 AI THERMAL SHIELD (LEARNING FUNCTION)

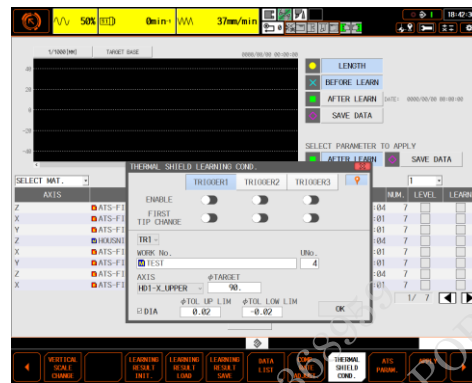
This function is designed to perform learning and optimize parameters according to the customer's factory environment and the operating pattern in order to ensure a high machining accuracy.

3-1 Learning Procedure

The dimensions of machined workpieces are entered in the NC unit and used for learning.

Setting of learning conditions

Set the conditions for collecting the data to be used for learning. In addition, specify whether to enable or disable the execution (trigger) of the collection of data under the set conditions.



<Reference>

3-3 Setting of Learning Conditions (THERMAL SHIELD LEARNING COND. Window)

Machining

Enable the trigger and execute machining. During the machining that meets the conditions set as learning conditions, the NC unit automatically collects the data to be used for learning.



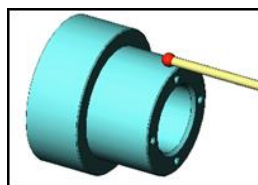
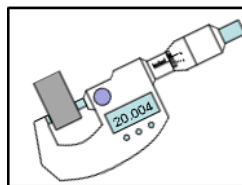
<Reference>

3-5 Examples of Application

Measurement

Measure the dimension of the machining portion set as a learning condition. The measurement can be performed in two ways: manual measurement and measurement using a measurement result output macro.

When machining is performed following a measurement, machining and measurement are repeated, but measurement does not necessarily need to be performed each time.



<Reference>

3-4-3 Measurement result output macro
3-5 Examples of Application

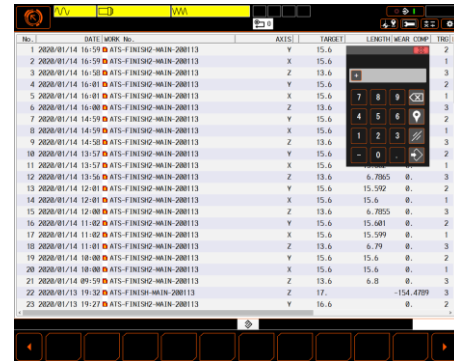
3

Ai THERMAL SHIELD (LEARNING FUNCTION)

Input of dimensions

Enter the measurement results.

The measurement results can be entered in two ways: manual input and input by loading a file.



No.	DATE	TIME	Meas. No.	AXIS	TARGET	LENGTH	WEAR	CONC.	THICK.
1	2020/01/14	16:59	ATS-FINISHQ-HALIN-200113	Y	15.6				2
2	2020/01/14	16:59	ATS-FINISHQ-HALIN-200113	X	15.6				1
3	2020/01/14	16:59	ATS-FINISHQ-HALIN-200113	Z	13.6				3
4	2020/01/14	16:01	ATS-FINISHQ-HALIN-200113	Y	15.6				2
5	2020/01/14	16:01	ATS-FINISHQ-HALIN-200113	X	15.6				1
6	2020/01/14	16:00	ATS-FINISHQ-HALIN-200113	Z	13.6				3
7	2020/01/14	14:59	ATS-FINISHQ-HALIN-200113	Y	15.6				2
8	2020/01/14	14:59	ATS-FINISHQ-HALIN-200113	X	15.6				1
9	2020/01/14	14:58	ATS-FINISHQ-HALIN-200113	Z	13.6				3
10	2020/01/14	13:59	ATS-FINISHQ-HALIN-200113	Y	15.6				2
11	2020/01/14	13:57	ATS-FINISHQ-HALIN-200113	X	15.6				1
12	2020/01/14	13:56	ATS-FINISHQ-HALIN-200113	Z	13.6	6.7865	0.		3
13	2020/01/14	12:01	ATS-FINISHQ-HALIN-200113	Y	15.6	15.592	0.		2
14	2020/01/14	12:00	ATS-FINISHQ-HALIN-200113	X	15.6	15.6	0.		1
15	2020/01/14	12:00	ATS-FINISHQ-HALIN-200113	Z	13.6	6.7855	0.		3
16	2020/01/14	11:02	ATS-FINISHQ-HALIN-200113	Y	15.6	15.601	0.		2
17	2020/01/14	11:02	ATS-FINISHQ-HALIN-200113	X	15.6	15.599	0.		1
18	2020/01/14	11:01	ATS-FINISHQ-HALIN-200113	Z	13.6	6.79	0.		3
19	2020/01/14	10:00	ATS-FINISHQ-HALIN-200113	Y	15.6	15.6	0.		2
20	2020/01/14	10:00	ATS-FINISHQ-HALIN-200113	X	15.6	15.6	0.		1
21	2020/01/14	09:59	ATS-FINISHQ-HALIN-200113	Z	13.6	6.8	0.		3
22	2020/01/13	19:32	ATS-FINISHQ-HALIN-200113	Z	17.	-154.4709			3
23	2020/01/13	19:27	ATS-FINISHQ-HALIN-200113	Y	16.6				2

<Reference>

3-4 Creation of Learning Data
(Learning Data List Display)

Learning

Execute learning using the entered learning data.

Check the learning results displayed in a graph and optimize parameters.



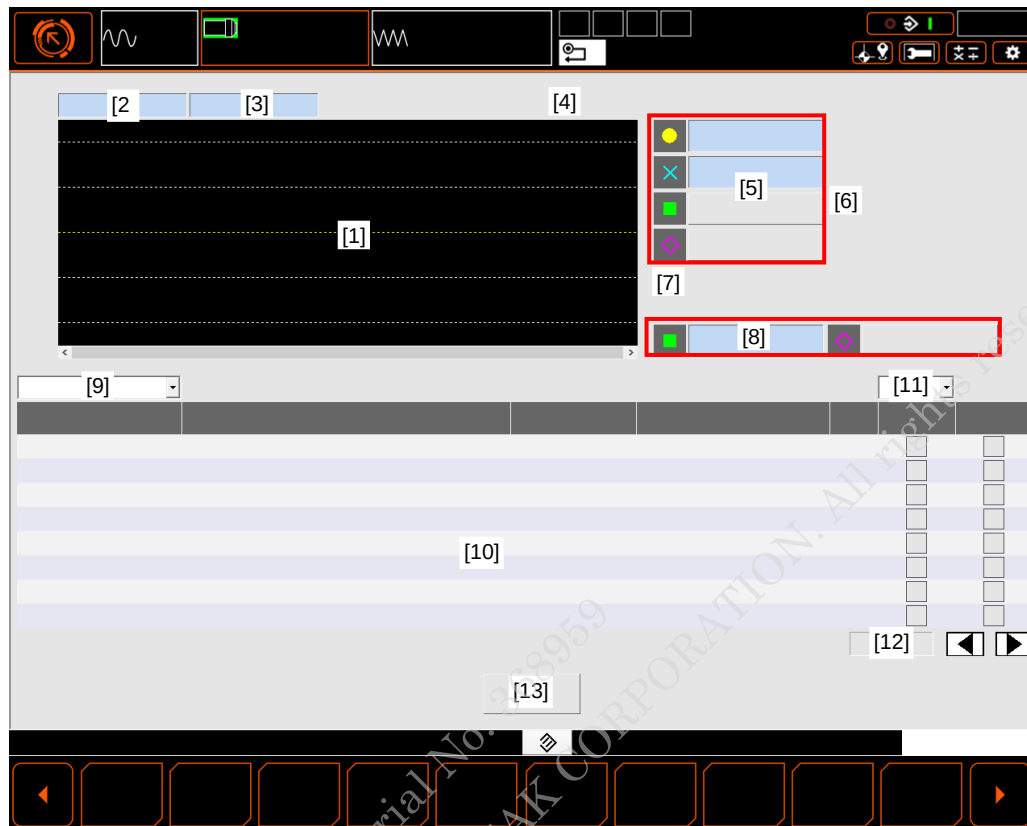
<Reference>

3-2 Ai THERMAL SHIELD LEARNING
Display

3-2 Ai THERMAL SHIELD LEARNING Display

Learning is performed using learning data.

1. Displayed data



2. Description of displayed data

No.	Item	Unit	Description
[1]	Dimension change graph	—	Displays the dimension entered as learning data and a graph of the machining accuracy before and after learning. ●: Dimension entered in the leaning data list display (LENGTH) ×: Reference dimension obtained by subtracting a wear compensation value and thermal displacement compensation value from LENGTH (BEFORE LEARN) ■: Dimension of learning results after simulation (AFTER LEARN) ◆: Results of a dimension simulation performed using the loaded learning results (SAVE DATA) ■: Tolerance set in the THERMAL SHIELD LEARNING COND. window Displayed only when the TARGET BASE button is enabled.
[2]	Diameter display switching button	—	Used to switch the display mode of the dimension change graph between radius and diameter values.
[3]	TARGET BASE button	—	Used to switch the display mode of the dimension change graph to the target value standard. When enabled: The center line of the graph is displayed in yellow. The data displayed in the graph is positioned on the basis of the target value. When disabled: The center line of the graph is displayed in white. The data displayed in the graph is positioned on the basis of a first product.
[4]	Date and time	Y/M/D/h/m	Displays the date and time that correspond to the position of the selected learning data.

No.	Item	Unit	Description
[5]	Graph display item change buttons	—	Used to change the data to be displayed in the graph.
[6]	Date and time of leaning	Y/M/D/h/m	Displays the date and time of obtaining the learning results indicated in the dimension change graph.
[7]	Loaded file name	—	Displays the name of the loaded file that contains the learning result data.
[8]	Parameter selection buttons	—	Used to select the parameters to be applied. Select AFTER LEARN or LOAD .
[9]	AXIS	—	Select the compensation axis for which learning is performed.
[10]	Learning pattern list	—	Displays the learning patterns used for learning. Remark : For details refer to "3. Contents of the learning pattern list."
[11]	Learning level	—	Used to select a learning level. A learning level can be selected from 5 options. In learning, priority is given to data at a higher learning level. If no learning level is selected, it is assumed to be level 1.
[12]	Page switch buttons	—	Used to switch pages.
[13]	LEARN button	—	Used to execute learning.

3. Contents of the learning pattern list

The learning pattern is created by extracting the learning data with the same conditions from the one registered in the leaning data list display. One learning pattern includes one first product (the first product field on the learning data list is ON). The learning pattern is separated by the first product and displayed as a different leaning pattern when there are multiple first products with the same condition.

Item	Unit	Description
AXIS	—	Displays the compensation axis for which learning is performed.
WNo.	—	Displays the work number. a b a: Identification of MAZATROL program or EIA/ISO program b: Work number
TARGET	mm (inch)	Displays the target value.
DATE	Y/M/D/h/m	Displays the date and time when the collection of learning data is started.
NUM.	—	Displays the number of the applicable data items.
LEVEL	—	Select whether or not to set a learning level. If selected, this item is marked with a check.
LEARN	—	Select whether or not to be used for learning. If selected, this item is marked with a check.

4. Display operation

VERTICAL SCALE CHANGE		LEARNING RESULT INIT.	LEARNINGR ESULT LOAD	LEARNINGR ESULT SAVE	DATA LIST	COMP. RATE ADJUST.	THERMAL SHIELD COND.		APPLY
1		2	3	4	5	6	7		8

<Menu functions>

1. VERTICAL SCALE CHANGE

Used to change the scale of the vertical axis of the dimension change graph.

2. LEARNING RESULT INIT.

Used to return the learning data to the factory setting.

3. LEARNING RESULT LOAD

Used to load the learning results.

Remark : For details refer to “3-2-2 Save and loading of the learning results.”

4. LEARNING RESULT SAVE

Used to save the learning results.

Remark : For details refer to “3-2-2 Save and loading of the learning results.”

5. DATA LIST

Used to move to the learning data list display.

Remark : For details refer to “3-4 Creation of Learning Data (Learning Data List Display).”

6. COMP. RATE ADJUST.

Used to move to the compensation factor adjustment display.

Remark : For details refer to Chapter 2 “Ai THERMAL SHIELD (COMPENSATION FACTOR ADJUSTMENT).”

7. THERMAL SHIELD COND.

Used to display the **THERMAL SHIELD LEARNING COND.** window.

Remark : For details refer to “3-3 Setting of Learning Conditions (THERMAL SHIELD LEARNING COND. Window).”

8. APPLY

Used to apply the learning results.

3-2-1 Learning procedure

This subsection describes the procedure for the operations to be performed after the entry of learning data.

- (1) Move to the learning display.
 - ➔ The created learning data is loaded and the learning patterns is displayed.
- (2) Select an axis to be compensated.
 - ➔ Only the learning patterns for the selected axis are displayed.
- (3) From the learning pattern list, tap the data to be used for learning.
 - ➔ The contents of the selected learning pattern are displayed in a graph.
- (4) Change the learning level if necessary.

Remark : Select the learning level using the learning level select button [11] and then place a check mark in the LEVEL check box for the intended learning pattern.
- (5) Tap the **[LEARN]** button.
 - ➔ Learning starts. When learning is completed, the machining accuracy after learning is displayed.
- (6) Tap the **[APPLY]** menu item.
 - ➔ The parameters are replaced with those optimized through learning.

Note 1: Learning patterns including the learning data with no dimension cannot be used for learning.

Note 2: The learning results cannot be applied during automatic operation.

Note 3: When the compensation axis (learning subject) is not selected, learning is not performed.

Note 4: If the amount of compensation is adjusted using the compensation factor adjustment function, the factor is automatically set to "1."

Note 5: Learning data collected more than 3 months ago from the current date cannot be used for learning.

Note 6: If 4 or more pieces of learning data are not collected within 24 hours, that learning pattern cannot be used for learning.

Note 7: A maximum of 2 learning patterns can be set for a single learning session. In addition, when 2 learning patterns are set, a different learning level cannot be applied to each of them.

3-2-2 Save and loading of the learning results

By saving the parameters optimized through learning, it becomes possible to compare the current learning results with past learning results and to load the parameters in another machine of the same specifications.

1. Save of the learning results

To save the learning results, follow the steps below.

- (1) Tap the [**LEARNING RESULT SAVE**] menu item.

→ The learning result save window appears.

- (2) Enter a file name.

Remark : A character string of up to 32 characters can be used as a file name. The characters and symbols that can be used for file names are: one-byte alphabetical characters A to Z and a to z, one-byte Arabic numerals, and one-byte symbols "+," "-" and ".".

- (3) Tap the [**OK**] button.

→ The learning results are saved.

2. Loading of the learning results

To load the learning results, follow the steps below.

- (1) Tap the [**LEARNING RESULT LOAD**] menu item.

→ The learning result loading window appears.

- (2) Select the file to be loaded.

- (3) Tap the [**OK**] button.

→ The learning results are loaded and displayed in the dimension change graph.

- (4) If the loaded learning data is applied to the machine, tap the [**SAVE DATA**] button and tap the [**APPLY**] menu for selection of parameters to be applied.

→ The parameter of the loaded learning result is applied.

3. Loading of the learning results in another machine of the same specifications

Parameters optimized through learning can be used in other machines of the same specifications. If an attempt is made to load inappropriate learning results, the alarm **505 CANNOT LOAD (MISMATCH)** will be triggered.

<Main criteria for determination>

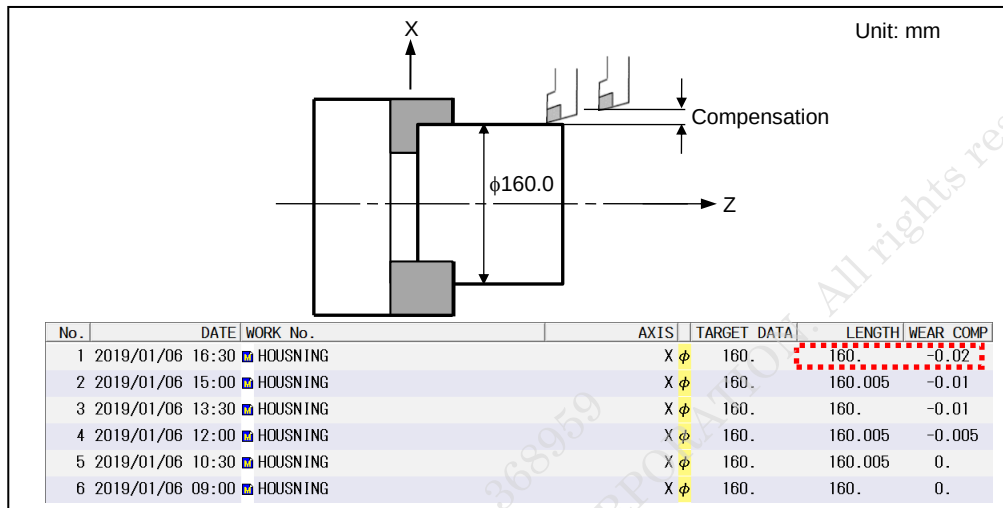
- Model name of the machine
- Whether or not the machine is equipped with a second turning spindle
- Whether or not the machine is equipped with a lower turret

Note : If parameters are loaded in another machine of the same specifications, the optimized parameters in the machine will be overwritten with the loaded ones. So, be sure to backup all important parameters before loading new ones.

3-2-3 Examples of learning

1. An example in which 6 products are manufactured between 9:00 AM and 16:30 PM

The diameter is between 160.000 and 160.005 mm and it can be seen that the diameter will satisfy the tolerance by entering wear compensation values. For example, in the case of product No. 1, a wear compensation value of -0.020 mm is entered and the diameter is 160.00 mm. So if no compensation is made for wear during machining, the diameter will be 160.20 mm, which is at the upper tolerance limit. During the manufacturing of products Nos. 1 through 6, the wear compensation value was changed 3 times, so it can be seen that extra work was done to enter the value manually.



To eliminate the need to manually enter a wear compensation value, optimize the parameters. In the learning window, select the learning data for the subject workpiece, check the dimension change graph and execute learning. After the learning has been completed, the learning results are displayed in the dimension change graph.

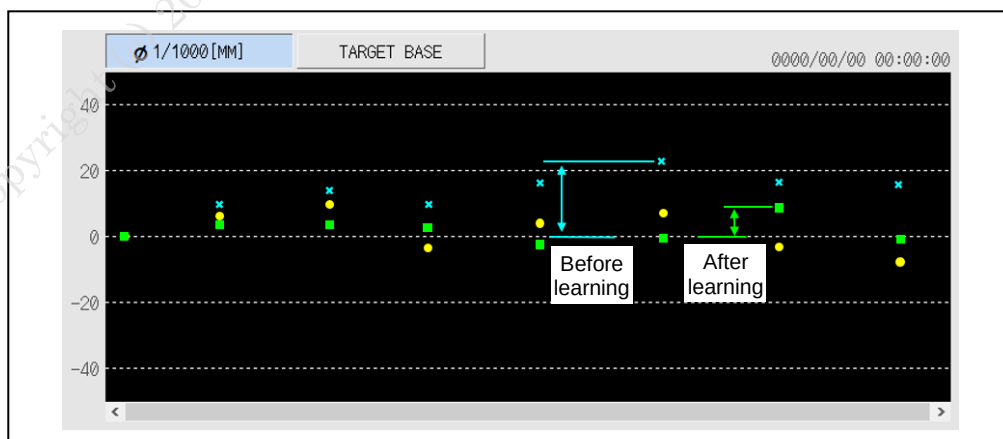
It can be seen that the dimensional change after learning is smaller than that before learning.

If there is a need to further improve the learning results, change the learning level and execute learning again.

If good learning results are obtained, execute the optimization of the parameters by tapping the **[APPLY]** menu item.

Power off the machine and check to see if the accuracy is improved on the next day.

<Display of the graph>



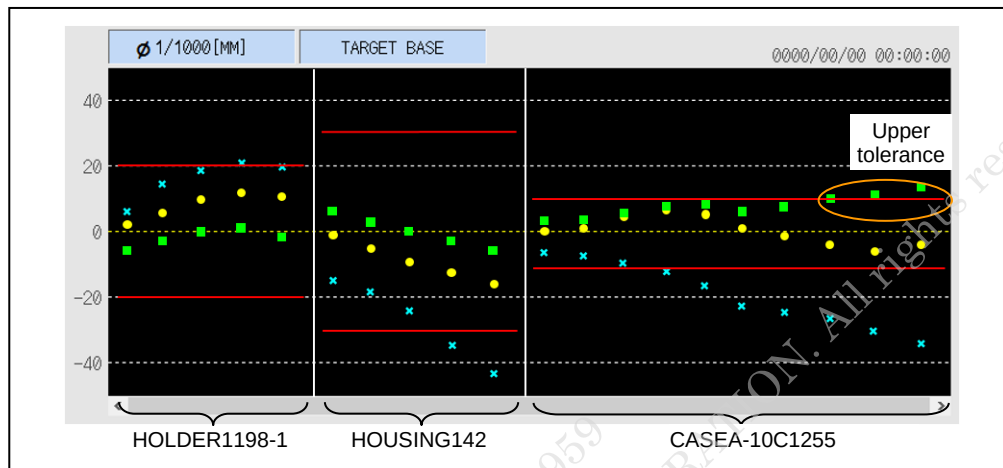
<Learning data>

AXIS	WNo.	TARGET	LEVEL	LEARN
X	HOUSNING	160		✓

2. An example in which 3 kinds of products are manufactured in 3 days

As a result of the learning performed without changing the learning level, the dimension change is reduced, but the dimension of the workpiece CASEA-10C1255 is at the upper tolerance limit. On the other hand, the other workpieces have sufficient margin to satisfy the tolerance.

<Display of the graph>



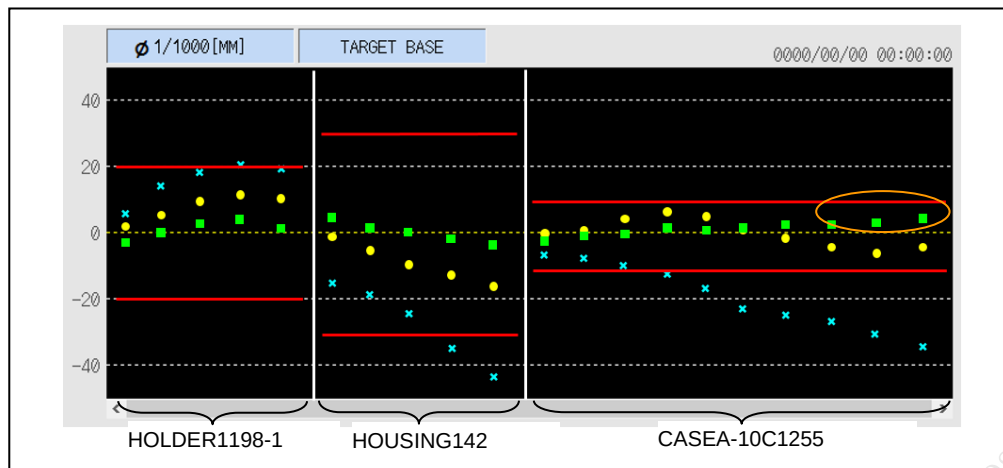
Raise learning level for the workpiece CASEA-10C1255 and perform learning again.

<Learning data>

			Learning level 5	
AXIS	WNo.	TARGET	LEVEL	LEARN
X	HOLDER1198-1	35		✓
X	HOUSING142	60		✓
X	CASEA-10C1255	50	✓	✓

It can be seen that good machining results have been obtained for the workpiece CASEA-10C1255 as well and the dimension is no longer at the upper tolerance limit.

<Display of the graph>



As good results have been obtained, execute the optimization of the parameters by tapping the **[APPLY]** menu item.

Power off the machine and check to see if the accuracy is improved on the next day.

Remark : Although the learning level can be set high for multiple learning data, better results can be obtained if the learning level is set high only for a particular piece of learning data.

3-3 Setting of Learning Conditions (THERMAL SHIELD LEARNING COND. Window)

Set the conditions for the collection of the data to be used for learning.

1. Displayed data

2. Description of displayed data

No.	Item	Unit	Description
[1]	TRIGGER buttons	—	Used to select the learning conditions to be set. A maximum of 3 learning conditions can be set.
[2]	Trigger ENABLE buttons	—	Used to enable or disable the collection of data under the set conditions. If some items are missing from the learning conditions, the collection of data cannot be enabled.
[3]	FIRST TIP CHANGE buttons	—	Enable this button when machining the first product or when performing machining for the first time after the change of tips.
[4]	Headstock/Turret switching	—	Used to switch the related headstock and turret.
[5]	WORK No.	—	Set a work number.
[6]	UNo./SNo.	—	Enter a unit number if a MAZATROL program is selected. Enter a sequence number if an EIA/ISO program is selected.
[7]	BNo.	—	Enter a block number if an EIA/ISO program is selected.
[8]	AXIS	—	Specify the compensation axis for which learning is performed.
[9]	TARGET	mm (inch)	Set the target dimension of the machining portion specified as a condition. Setting range: -99999.9999 to 99999.9999 (metric) -9999.99999 to 9999.99999 (inch)
[10]	DIA checkbox	—	Check this when entering a diameter value as the target value or target tolerance.
[11]	TOL UP LIM TOL LOW LIM	mm (inch)	Set a tolerance for the machining portion specified as a condition. Upper tolerance setting range: -99.9999 to 99.9999 (metric) -9.99999 to 9.99999 (inch) Lower tolerance setting range: -99.9999 to 99.9999 (metric) -9.99999 to 9.99999 (inch)
[12]	OK button	—	Used to save the settings.

3. How to display

The **THERMAL SHIELD LEARNING COND.** window can be displayed in the following 3 ways.

- Using the menu displayed when in the memory operation mode the menu selection button in the **POSITION** display is tapped.
- Using the menu in the Ai **THERMAL SHIELD LEARNING** display.
- Using the menu in the Ai **THERMAL SHIELD LEARNING DATA** display.

4. Precautions

- Learning subjects that can be selected differ depending on the machine specifications.
- The **FIRST TIP CHANGE** button is automatically deactivated after the completion of the machining of the first product.
- The collection of data should be disabled when performing machining on a trial basis.
- If the specified work number or unit number does not exist, no data will be collected.
- If the period during which the specified learning conditions are satisfied is too short, data will not be collected.
- If the sequence of the selected sequence No. calls an EIA/ISO subprogram, multiple data may be collected under the same conditions.
- For following units in which the rough machining and finish machining are distinguished in the MAZATROL program, data is not collected in the rough machining.
 - Line machining unit
 - Face machining unit
 - Turning machining unit
 - C-axis line machining unit
 - C-axis face machining unit
- If one of the following units of a MAZATROL program is selected and priority numbers are assigned in the unit, multiple data may be collected under the same conditions.
 - Point machining unit
 - C-axis point machining unit
- When the number of learning data reaches 2000, the alarm **2658 THERMAL SHIELD LEARN DATA OVER** occurs and data is not collected any more. Delete unnecessary learning data in the learning data list display.

3-4 Creation of Learning Data (Learning Data List Display)

Enter a dimension to create the data to be used for learning.

1. Displayed data



2. Description of displayed data

No.	Item	Description
[1]	Learning data No.	Displays the numbers of learning data.
[2]	LEARN DATA	Displays various kinds of learning data. Remark : For details, refer to "3. Contents of learning data."

3. Contents of learning data

Item	Unit	Description
DATE	Y/M/D/h/ m	Displays the date and time when machining was performed.
WORK No.	—	Displays the work number. a b a: Identification of MAZATROL program or EIA/ISO program b: Work number
AXIS	—	Displays the compensation axis for which learning is performed.
TARGET	mm (inch)	Displays the target value.
LENGTH	mm (inch)	Displays the entered dimensions. Setting range: -99999.9999 to 99999.9999 (metric) -9999.99999 to 9999.99999 (inch)
WEAR COMP	mm (inch)	Displays the amount of wear compensation.
TRG	—	Displays the numbers of learning conditions.
First product	—	Displays the status of the FIRST TIP CHANGE button.
UNo./SNo.	—	Displays the unit number or the sequence number.
BNo.	—	Displays the block number. This field is blank if a MAZATROL program is used for machining.
Upper tolerance limit	mm (inch)	Displays the upper tolerance limit.
Lower tolerance limit	mm (inch)	Displays the lower tolerance limit.

Remark 1: The data set in the learning condition setting process are automatically entered except for dimensions (LENGTH).

Remark 2: Tapping the item name sorts the display order of that item in ascending or descending order. When displaying the **Learning Data List** display again after switching to other displays, the display order returns to the date order.

4. Display operation



<Menu functions>

1. INPUT LENGTH

Used to enter a dimension. Enable this menu item to enter a dimension.

2. DATA ERASE

Used to delete the created learning data.

Remark : For details, refer to “3-4-2 Deletion of learning data.”

3. LOAD LENGTH

Used to read dimensions from a file.

Remark : For details, refer to “3-4-1 Loading of dimension data from a file.”

4. DATA OUTPUT

Used to output the learning data to a CSV file.

5. THERMAL SHIELD COND.

Used to display the **THERMAL SHIELD LEARNING COND.** window.

Remark : For details refer to “3-3 Setting of Learning Conditions (THERMAL SHIELD LEARNING COND. Window).”

6. LEARN

Used to move to the learning display.

Remark : For details, refer to “3-2 Ai THERMAL SHIELD LEARNING Display.”

5. Precautions

- A maximum of 2000 pieces of learning data can be registered.
- Only the data item of dimension (LENGTH) can be edited. Other data items cannot be edited.
- If the specified dimension or target value does not match the actual machining, no learning effect may be obtained.
- The learning data generated during automatic operation is applied at the timing of moving to the learning data list display after the automatic operation ends.

3-4-1 Loading of dimension data from a file

From a file, load the dimension data to be entered as learning data.

Dimension data can be loaded from the following two types of files.

- File generated by a measurement result output macro program
- External file (CSV file)

1. Displayed data

2. Description of displayed data

No.	Item	Description
[1]	MEASUREMENT MACRO button	Used to load dimension data from the file generated by a measurement result output macro program.
[2]	EXTERNAL FILE button	Used to load dimension data from a CSV file.
[3]	Folder name	Displays the name of the folder that contains the file to be loaded.
[4]	File name	Displays the name of the file to be loaded.
[5]	BROWSE button	Used to specify the file to be loaded.
[6]	EXECUTE button	Used to execute the loading.

Note : The folder name field, the file name field and the browse button are active only when an external file is loaded.

3. Loading of a file generated by a measurement result output macro program

A. How to generate a file

To generate a file, create a measurement macro for in-process measurement and execute it during machining.



- For creation of macro programs refer to “3-4-3 Measurement result output macro.”

B. Loading of the file

- (1) Tap the **[LOAD LENGTH]** menu item.
→ The dimension loading window appears.
- (2) Tap the **[MEASUREMENT MACRO]** button.
- (3) Tap the **[EXECUTE]** button.
→ Dimension data is loaded.

Note 1: If the file to be loaded is not found, the alarm **1441 FILE NOT FOUND** will be triggered.

Note 2: If there is an error in the format of the data to be loaded, the alarm **401 ILLEGAL FORMAT** will be triggered.

Note 3: If the dimension of the same machining portion was measured multiple times in a single machining process, the dimension measured at the time closest to the time when machining was executed will be loaded.

Note 4: If the data that has been read once is read again, the alarm **2057 NUMBER ALREADY REGISTERED** occurs.

4. Loading a CSV file

If the dimensions of a workpiece are measured outside the machine, they can be saved in a CSV file and read as learning data.

A. How to create a file

Enter the dimension that corresponds to the number of the learning data. The dimension is entered where the number of the learning data in the learning data list window agrees with the number written on the left side of a comma in a CSV file.

1. File format

A,B

- A: Learning data number
- B: Dimension

2. Example of notation

If the dimension 123.4 mm is to be entered for learning data No. 10.

CSV file

10,123.4

B. Loading a file

- (1) Tap the **[LOAD LENGTH]** menu item.
→ The dimension loading window appears.
- (2) Tap the **[EXTERNAL FILE]** button.
- (3) Tap the **[BROWSE]** button.
→ The folder browsing window appears.
- (4) Select the CSV file to be loaded.
- (5) Tap the **[EXECUTE]** button.
→ The dimension data is loaded.

Note 1: When loading the file, disable all the triggers in the **THERMAL SHIELD LEARNING COND.** window. If there is any enabled trigger, it is possible that new learning data is added and the number in the learning data list is changed.

Note 2: Execute loading after checking that the learning data No. in the learning data list to be loaded matches the learning data No. described in the CSV file. If the dimension to be entered does not match the learning data, the learning can be not executed correctly.

Note 3: If the file to be loaded is not found, the alarm **1441 FILE NOT FOUND** will be triggered.

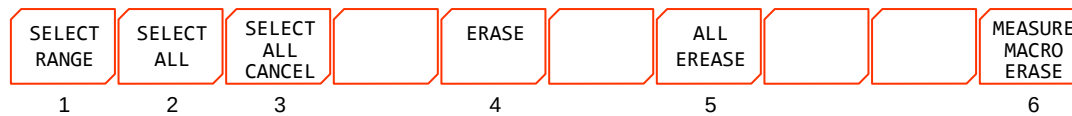
Note 4: If one of the following errors is found in the format of the file to be loaded, the alarm **401 ILLEGAL FORMAT** will be triggered.

- The learning data number entered is 0 or lower, or it is larger than 2000.
- The learning data number entered is already used for other dimension data.
- The format contains something that should not to be used in it.

Note 5: If the file name or file path entered is longer than 256 characters, the alarm **2039 FILE NAME TOO LONG** will be triggered.

3-4-2 Deletion of learning data

To delete learning data, tap the **[DATA ERASE]** menu item.



<Menu functions>

1. SELECT RANGE
Used to delete learning data while specifying the starting and ending points of the portion to be deleted.
2. SELECT ALL
Used to select all the data.
3. SELECT ALL CANCEL
Used to deselect all the data.
4. ERASE
Used to delete the selected data.
To delete the data, tap the **[OK]** button in the message box that appears.
5. ALL ERASE
Used to delete all the data.
6. MEASURE MACRO ERASE
Delete the data created by a measurement result output macro program.
To delete the data, tap the **[OK]** button in the message box that appears.

- When commanding in the overwriting mode, the previous measured result is deleted at the open command (POPEN) execution. Therefore, load dimensions for each measurement.
- When commanding in the editing mode, load dimensions after all measurements are completed.

The same data cannot be loaded again. Therefore, if there is any data that has been loaded already, delete it with the **[MEASURE MACRO ERASE]** menu before measurement.

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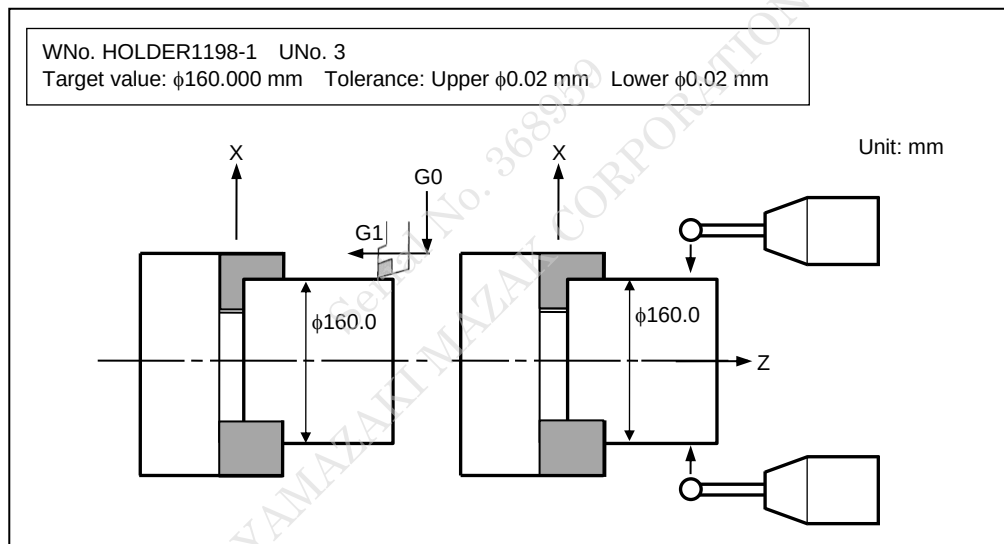
3-5 Examples of Application

1. The outside diameters of workpieces machined by turning decreased by 0.02 mm between morning and evening.

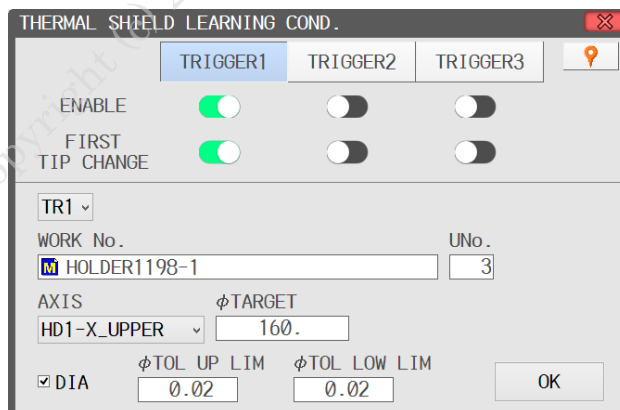
- Machine model: INTEGREX
- Machining: Finish turning of the HD1-side outer periphery ($\phi 160.00$ mm) using the upper turret
- Likely cause: The outside diameter ($\phi 160.00$ mm) decreased gradually, so this is likely caused by thermal displacement in the X-axis direction.
- Learning subject: HD1-side X-axis of upper turret
- Dimension: The outside diameter of the machined workpiece is measured on both sides, using a workpiece measurement unit.

Set learning conditions and perform machining. After the completion of the machining, data other than dimensions are displayed in the learning data list display that appears. Load the dimension (diameter) data in the blank dimension field from the file generated by a measurement result output macro and after that, perform learning.

A. Machining and measurement



B. Setting the learning conditions



C. MAZATROL program, using a workpiece measurement unit

A measurement is performed in the machine after the completion of the machining of the portion specified as a learning condition.

Example of a program

UNo.	UNIT		PART	CPT-X		CPT-Z		FIN-X		FIN-Z				
3	BAR		OUT	200.		0.		0.4		0.05				
SNo.	TOOL		NOM.		No.	PAT.		DEP-1	DEP-2/NUM.		DEP-3	FIN-X	FIN-Z	
R 1	GENERAL	OUT	1.	A		0		5.						
F 2	GENERAL	OUT	2.	B								0.	0.	
FIG	PTN		S-CNR		SPT-X		SPT-Z		FPT-X		FPT-Z		F-CNR/\$	R/th
1	LIN		C 1.						160.		100.			

UNo.	UNIT	COMPENSATE	OFS-TOOL			COMP. DATA		SNS-TOOL		No.	INTERVAL		OUTPUT		
3	WORK MES	NO						TOL SENS	5.	A	1		0		
SNo.	PTN	SPT-X	SPT-Y	SPT-Z	FPT-X	FPT-Y	FPT-Z	T LIM+	T LIM-	BASE			DIR.		
1	OUTER X	0	160.		-25.				0.02	-0.02					↗Z

UNo.	UNIT	WORK No.			\$	REPEAT	No.		
9	SUB PRO	DPRNTOUT							

← Machining unit for the portion specified as a condition

← Workpiece measurement unit

WNo. DPRNTOUT

```
G90G94
#550=#58075
POPEN ,0000 ,1
DPRNT[#3011[80]*#3012[60]*#550[44]*1]
PCLOS, 0
```

D. EIA/ISO program, using measurement macro (workpiece measurement pattern)

Example of a program

...
G96S1500M03
G0X164.0Z2.M8
X160.
N100
G1Z-50.0F0.1
...
...
G136X101.Z-10.I2.J-2.K5.U90.V0.W10.Q13
#550=#58075
POPEN ,0000 ,1
DPRNT[#3011[80]*#3012[60]*#550[44]*1]
PCLOS, 0
...
...

↔ Machining of the portion specified as a condition

↔ Measurement

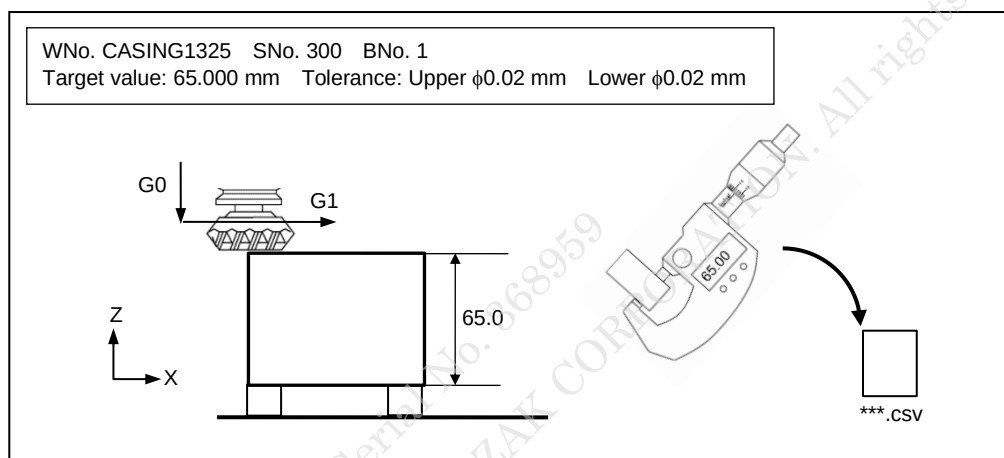
↔ Output of the measurement result

2. The thickness of machined workpieces changed by 0.03 mm between morning and evening.

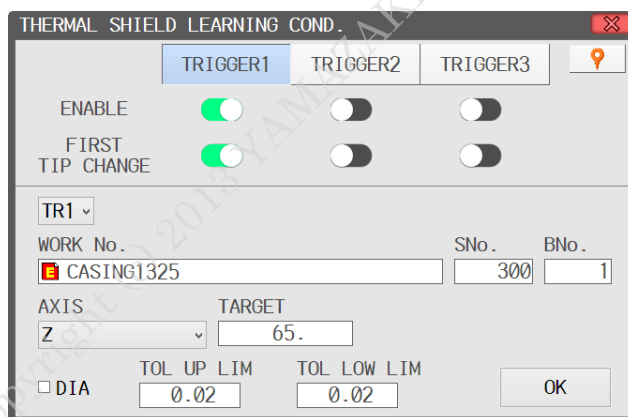
- Machine model: VARIAXIS
- Machining: Milling of the upper surface
- Likely cause: The thickness in the Z-axis direction changed, so this is likely caused by thermal displacement in the Z-axis direction.
- Learning subject: Z-axis
- Dimension: Measure the thickness, using an external measurement device, and save the measurement result in a CSV file.

Set learning conditions and perform machining. After a total of 6 workpieces are machined, data other than dimensions are displayed in the learning data list display that appears. Load the dimension (height) data in the blank dimension filed from the CSV file and after that, perform learning.

A. Machining and measurement



B. Setting the learning conditions



C. Creation of learning data

No.	DATE	WORK No.	AXIS	TARGET DATA	LENGTH	WEAR COMP
1	2019/01/06 16:30	CASING1325	Z	65.		0.
2	2019/01/06 15:00	CASING1325	Z	65.		0.
3	2019/01/06 13:30	CASING1325	Z	65.		0.
4	2019/01/06 12:00	CASING1325	Z	65.		0.
5	2019/01/06 10:30	CASING1325	Z	65.		0.
6	2019/01/06 09:00	CASING1325	Z	65.		0.

Loading an external file

1,64.980
 2,64.980
 3,64.985
 4,64.990
 5,65.005
 6,65.010

No.	DATE	WORK No.	AXIS	TARGET DATA	LENGTH	WEAR COMP
1	2019/01/06 16:30	CASING1325	Z	65.	64.98	0.
2	2019/01/06 15:00	CASING1325	Z	65.	64.98	0.
3	2019/01/06 13:30	CASING1325	Z	65.	64.985	0.
4	2019/01/06 12:00	CASING1325	Z	65.	64.99	0.
5	2019/01/06 10:30	CASING1325	Z	65.	65.005	0.
6	2019/01/06 09:00	CASING1325	Z	65.	65.01	0.

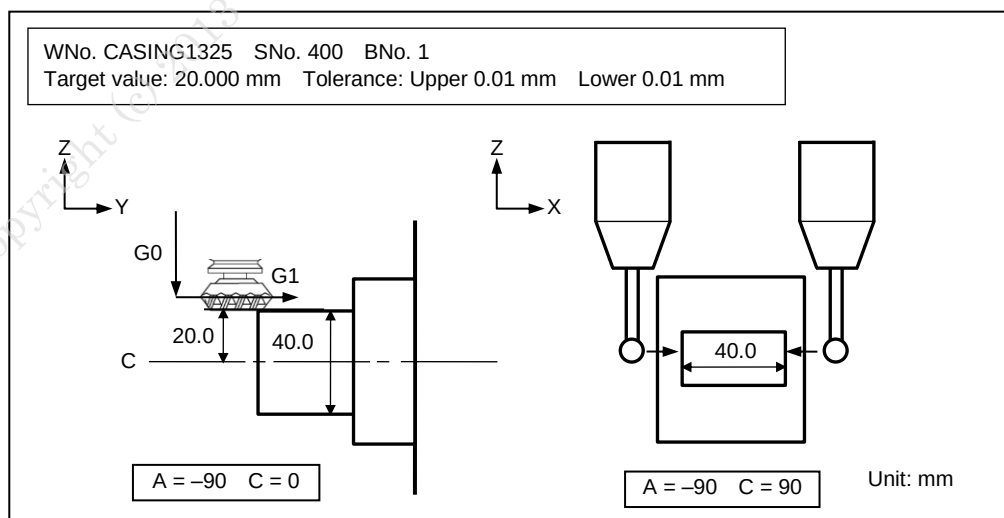
Check to see that the dimension is read correctly and after that, perform learning.

3. The width between machined two faces changed by 0.03 mm between morning and evening.

- Machine model: VARIAXIS
- Machining: Milling with A-axis at -90° , C-axis at 0° and A-axis at -90° , and C-axis at 180°
- Likely cause: The thickness in the Z-axis direction changed, so this is likely caused by thermal displacement in the Z-axis direction.
- Learning subject: Z-axis
- Dimension: Distance between the machined two faces of workpieces, measured inside the machine

Set learning conditions and perform machining and measurement. The measurement of the dimension in the machine is performed with the A-axis at -90° and the C-axis at 90° .

A. Machining and measurement



B. Setting the learning conditions

C. Creation of learning data

As with the example of application in 1, after the completion of the machining of the portion specified in the **THERMAL SHIELD LEARNING COND.** window, perform a measurement in the machine and execute the measurement result output macro. After the collection of learning data has been completed, load the dimension (width) data onto the learning data list display and after that, perform learning.

In this case, half of the measured value is entered as the dimension.

No.	DATE	WORK No.	AXIS	TARGET	DATA	LENGTH	WEAR	COMP
1	2019/01/06 16:30	CASING1325	Z	20.		19.987	0.01	
2	2019/01/06 15:00	CASING1325	Z	20.		19.988	0.01	
3	2019/01/06 13:30	CASING1325	Z	20.		19.995	0.006	
4	2019/01/06 12:00	CASING1325	Z	20.		20.	0.002	
5	2019/01/06 10:30	CASING1325	Z	20.		20.	0.	
6	2019/01/06 09:00	CASING1325	Z	20.		20.	0.	

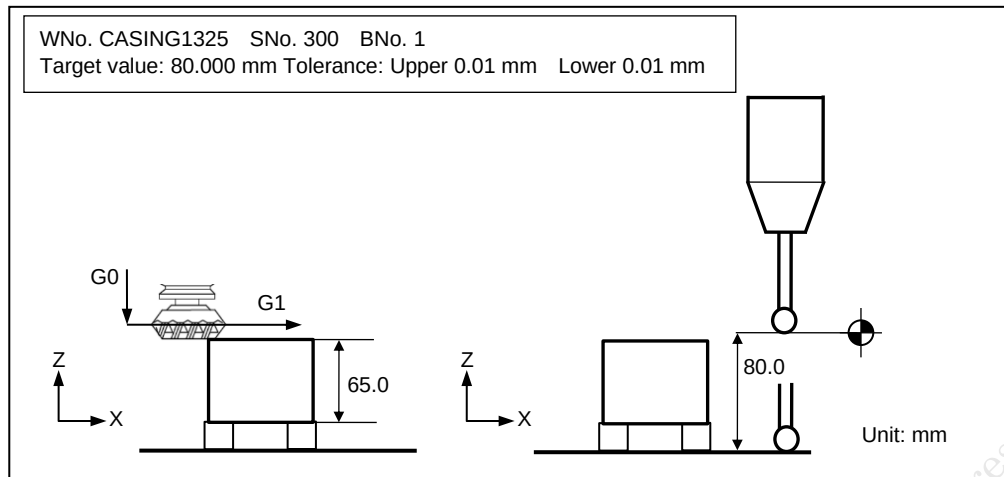
Check to see that the dimension is read correctly and after that, perform learning.

4. The thickness of machined workpieces changed by 0.03 mm between morning and evening. Learning data was collected by performing measurements at a fixed point.

- Machine model: VARIAXIS
- Machining: Milling of the upper surface
- Likely cause: The thickness in the Z-axis direction changed, so this is likely caused by thermal displacement in the Z-axis direction.
- Learning subject: Z-axis
- Dimension: Length from the program zero point to the upper surface of the table

Let us suppose that the thickness (65 mm) of workpieces machined by milling varies in accuracy. The more the Z-axis is displaced in the negative direction by heat, the more the thickness of a machined workpiece decreases. So, prepare a measurement macro to measure the length from the program zero point to the upper surface of the table, as shown in the figure below. The more the Z-axis is displaced in the negative direction, the more the length (80 mm) decreases. So, the length measured in the machine is learned instead of the change in the dimensions of machined workpieces.

Set the sequence number and block number of the execution of the measurement macro in the learning conditions. After the collection of learning data has been completed, load the measured dimension (length) data onto the learning data list display and after that, perform learning.

A. Machining and measurement**B. Setting the learning conditions**

THERMAL SHIELD LEARNING COND.

TRIGGER1 TRIGGER2 TRIGGER3

ENABLE ☒ ☐ ☐

FIRST TIP CHANGE ☒ ☐ ☐

TR1

WORK No. SNo. BNo.

CASING1325 300 1

AXIS TARGET

Z 80.

☐ DIA TOL UP LIM TOL LOW LIM

0.01 0.01

OK

C. EIA/ISO program, measurement macro

Example of program

```

G54.1P300
G0X0.Y0.
G0Z0.
#580=#5043(Z)
G04X1.0
Z=-75.
N300
G31G1Z-50.F50.
G04X1.0
#581=#5043(Z)
#582=#580-#581
G0Z0.
POpen,0000,1
DPRNT[#3011[80]*#3012[60]*#582[44]*1]
PCLOS,0

```

↔ After the spindle is positioned at the program zero point, its coordinates are read.

↔ A skip signal is input and the workpiece coordinate at the time when the machine stopped feeding is read.

↔ The length from the program zero point to the upper surface of the table is calculated.

↔ The measurement result is output.

After the collection of learning data has been completed, load the measured dimension (length) data onto the learning data list display and after that, perform learning.

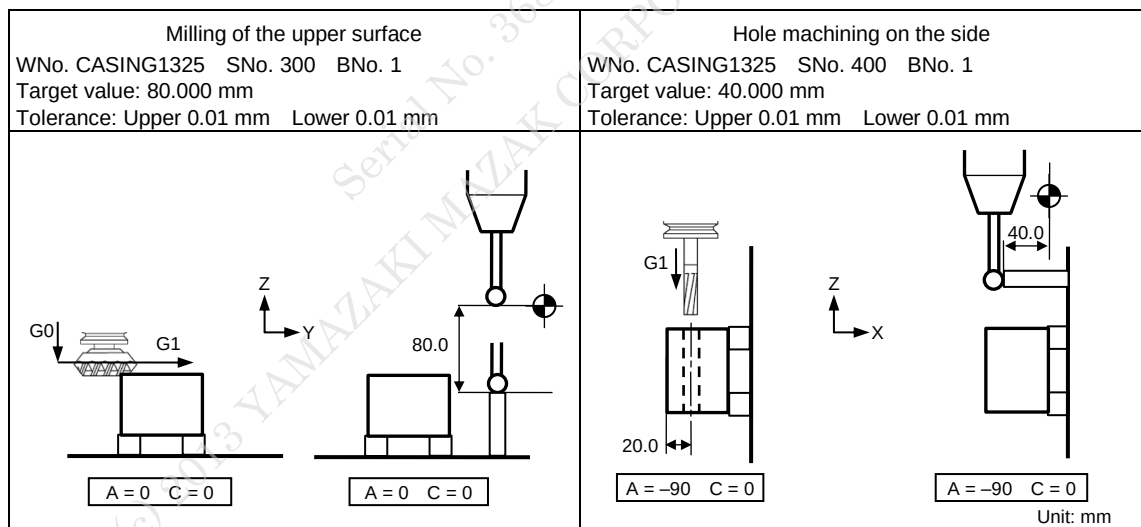
5. The dimension of machined workpieces changed by 0.03 mm between morning and evening. Learning data was collected by performing measurements at a fixed point.

- Machine model: VARIAXIS
- Machining: Milling of the upper surface and then hole machining on the side
- Likely cause: This is likely caused by thermal displacement in the Z- and Y-axis directions.
- Learning subject: Z-axis, Y-axis
- Dimension: Length from the program zero point to the upper surface of the fixture

Let us suppose that the dimension (20 mm) between the upper surface machined by milling and the hole on the side varies in accuracy. The accuracy of the machining of the upper surface is affected by thermal displacement in the Z-axis direction and that of the position of the hole on the side is affected by thermal displacement in the Y-axis direction. So, to stabilize the dimension at 20 mm, the displacement in both the Z- and Y-axis directions need to be learned. As with the example of application in 4, the length measured in the machine, using a touch sensor, is to be learned. Set the sequence number and block number of the execution of the measurement macro in the learning conditions. After the collection of learning data has been completed, load the measured dimension (length) data onto the learning data list display and after that, perform learning.

The closer the position of the machine at the time of machining and that at the time of a measurement at a fixed point, the higher the learning effect may increase.

A. Machining and measurement



B. Setting the learning conditions

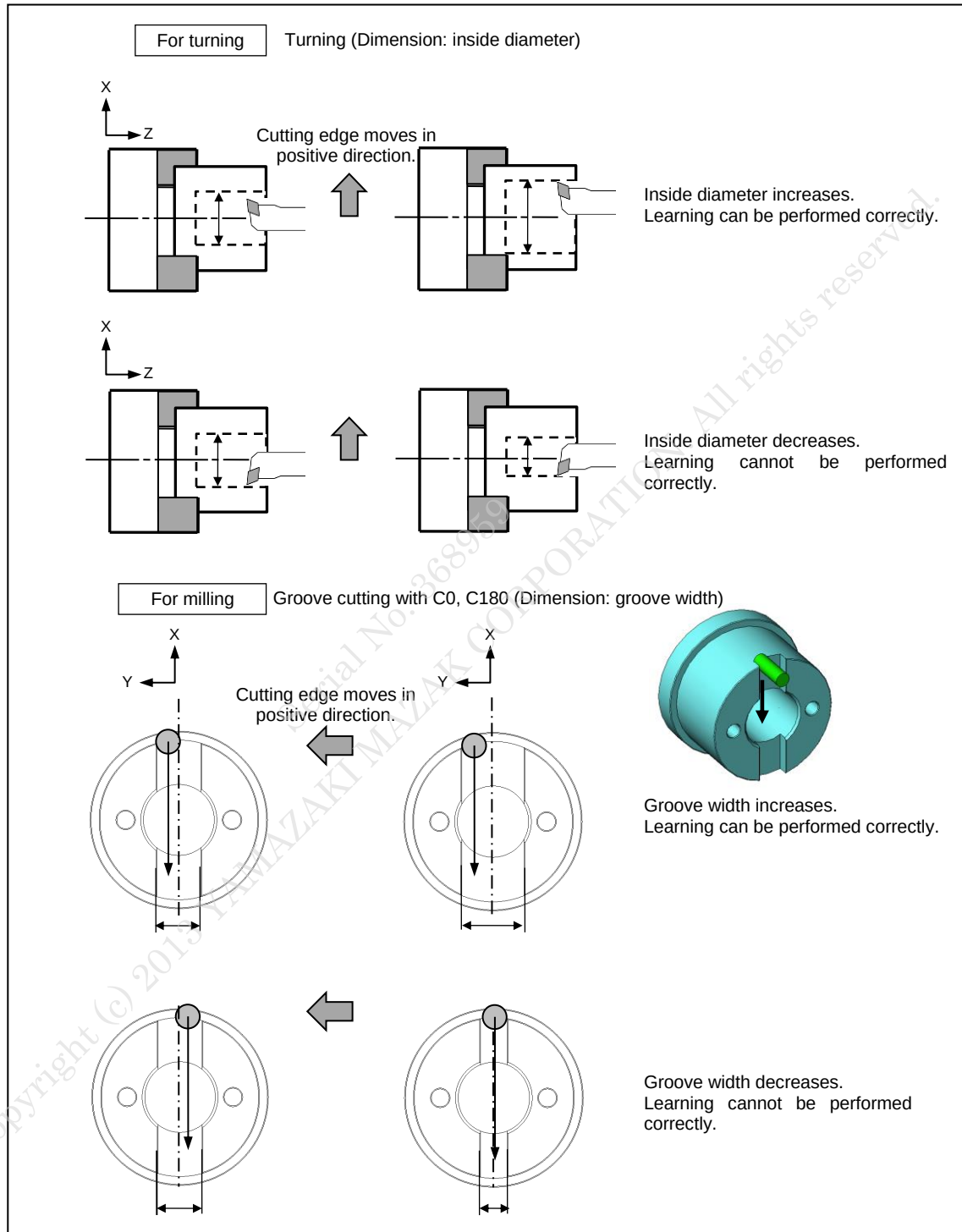
Milling of the upper surface	Hole machining on the side
THERMAL SHIELD LEARNING COND.	
TRIGGER1 TRIGGER2 TRIGGER3	
ENABLE FIRST TIP CHANGE	
TR1 WORK No. CASING1325 SNo. 300 BNo. 1	
AXIS Z TARGET 80.	
TOL UP LIM 0.01 TOL LOW LIM 0.01 OK	
TR1 WORK No. CASING1325 SNo. 400 BNo. 1	
AXIS Y TARGET 40.	
TOL UP LIM 0.01 TOL LOW LIM 0.01 OK	

3-6 Precautions

1. This function is not designed to guarantee accuracy.
2. Learning involves a change in the amount of thermal displacement compensation and, therefore, has a direct effect on alignment and machining dimensions. When a workpiece is machined for the first time after learning, due allowance should be made for dimensional errors.
3. The change in compensation factor is used for all tools used for the machining concerned.
4. The amount of compensation set with the thermal displacement compensation function is not incorporated in the POSITION and MACHINE values on the **POSITION** display, which means that learning does not cause any change on the **POSITION** display.
5. When using this function, determine the change in the amount of thermal displacement compensation on a daily basis and perform learning for the next day's operation. Adjustments and learning should not be performed many times a day.
6. If the time in the NC unit is changed, learning cannot be performed correctly. If the time is changed, the collection of learning data should not be performed for 24 hours after that.
7. If Ai Thermal Shield is down due to a wire break in the thermal sensor or for any other reason, do not try to collect data or execute learning, using the learning conditions. If this is the case, contact Mazak Technical Center or Technology Center.

3-7 Restrictions

Learning cannot be performed correctly if machining in which the dimension decreases as the cutting edge moves in the positive direction of the machine coordinate system is set as a learning condition.





Handwriting practice area with horizontal dashed lines.

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4 TROUBLESHOOTING

This chapter explains what to do if you encounter a problem.

4-1 Automatic Operation Locked due to a Thermal Displacement Compensation Error

Due to a wire break in the thermal sensor or for any other reason, a thermal displacement compensation error occurs.

In this case, contact Mazak Technical Center or Technology Center.

If automatic operation needs to be performed urgently, disable the thermal sensor.

<Procedure>

(1) Tap the  button on the NC screen and then tap .

(2) Tap **[MAINTENANCE]** and then tap **[THERMO SENSOR DISABLE]**.

Note 1: When the thermal sensor is disabled, the thermal displacement compensation function does not work.

Therefore, it may be difficult to maintain the accuracy of the machine.

Note 2: When the thermal sensor is disabled, do not use the compensation factor adjustment function or the learning function.



Area with horizontal dashed lines for writing or notes.

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